Chance Constrained Optimization with Kernel Density Estimation

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1 The Problem of Optimization Under Uncertainty

Every optimization model is a promise. It promises that if the decision it recommends is implemented, the constraints of the system will hold and the objective will be as good as the mathematics claims. That promise is honest only if the numbers inside the model are honest, and in almost every real deployment they are not. The demand a grid operator schedules against is a forecast, not a fact. The thrust a lander commands is corrupted by disturbances no sensor fully resolves. The returns a portfolio is built on are realizations of a process nobody has ever observed in closed form. This chapter establishes precisely what goes wrong when uncertainty is ignored, why the two classical remedies, ignoring it and armoring against the worst case, both fail in practice, and why the chance constraint, a constraint required to hold with a prescribed probability, is the correct middle ground. It then exposes the three structural reasons chance constraints are computationally hard, builds the case for estimating the relevant probability distributions from data rather than assuming them, and introduces the single conceptual shift on which the entire thesis rests: moving the estimation problem from the space of the uncertainty to the space of the constraint residual. The chapter closes by fixing the scope, the anchor literature, and the research questions of the thesis.

1.1 Where Deterministic Optimization Falls Short

1.1.1 The mean-value approach and the flaw of averages

The standard nonlinear program

$$\underset{x\in X}{\text{minimize}}\; f(x) \quad \text{subject to} \quad g(x,\xi) \le 0 \tag{1}$$

treats the parameter vector ξ as a known constant. Decades of solver development have made problems of the form Eq. (1) routinely solvable at scale, and this success has had a quiet side effect: the deterministic template is applied by default, even when ξ is plainly a random quantity. The usual workaround is to replace ξ with a point estimate, most often its mean or a nominal forecast, and solve the resulting deterministic problem. This is the mean-value approach, and it fails in a way that is systematic rather than occasional.

The failure mechanism is worth stating carefully, because it is the engine behind every example in this subsection. An optimizer pushes its solution to the boundary of the feasible region; that is what optimality means whenever a constraint is binding. If the constraint $g(x, \bar{\xi}) \leq 0$ is active at the mean-value optimum $x^\star_{\mathrm{mv}}$, then the residual $g(x^\star_{\mathrm{mv}}, \xi)$ sits exactly at zero when ξ equals its mean. The moment ξ is allowed to fluctuate around that mean, roughly half of its realizations push the residual positive and the constraint is violated. The optimizer does not merely tolerate risk; it actively seeks out the configuration in which risk is maximal, because that configuration is where the objective is best. Optimization and uncertainty are adversaries by construction, a phenomenon known in the planning literature as the flaw of averages (Savage, 2009): a plan built on average inputs is not an average plan, it is a fragile one.

1.1.2 Four domains where uncertainty bites

Four application domains make this concrete. They are chosen deliberately, because each reappears later in the thesis as a benchmark or a case study, and because together they span the distributional pathologies that motivate the nonparametric approach of **??**.

Autonomous lunar landing. A powered-descent guidance problem asks for a thrust profile that brings a lander from orbit to a designated site while respecting bounds on velocity, glide slope, and fuel. The dynamics are integrated forward under thrust perturbations and imperfectly known gravitational and terrain parameters. Keil u. a. (2021) show that disturbance distributions in this setting are naturally non-Gaussian; a mixture arises, for example, when a thruster operates in one of two regimes. A trajectory optimized for nominal thrust grazes the velocity constraint at touchdown, so under perturbation the lander exceeds its safe impact speed on a large fraction of descents. There is no second attempt at a lunar landing. Chance-constrained formulations of exactly this problem are the application anchor of the thesis (Keil u. a., 2020, 2021; Blackmore u. a., 2010).

Energy dispatch under renewable generation. A system operator commits generation a day ahead against uncertain demand and uncertain wind and solar output. Forecast errors for renewable infeed are well documented to be skewed and heavy-tailed rather than Gaussian (Hodge u. Milligan, 2011; Bienstock u. a., 2014). A dispatch that balances the forecast exactly is short of energy whenever net demand lands above its forecast, which by construction happens with probability near one half, and reserve activations, load shedding, or price spikes follow. This domain has produced some of the strongest data-driven chance-constraint literature (Ciftci u. a., 2019; Wu u. a., 2021; Wu u. Kargarian, 2023).

Chemical process control. A reactor recipe is optimized for yield subject to temperature and purity limits, but the composition of the feed material varies from batch to batch. Running the process at the deterministic optimum places the temperature constraint on the boundary, and ordinary feedstock variation then produces off-specification batches at a steady rate. Process systems engineering was, in fact, the field where data-driven kernel-smoothed chance constraints were first formulated (Calfa u. a., 2015).

Portfolio allocation. An investor maximizes expected return subject to a cap on the probability of a large loss. Asset returns exhibit heavy tails and volatility clustering; these are among the most robust stylized facts in empirical finance (Cont, 2001). A portfolio optimized as if returns were fixed at their historical means concentrates in the highest-mean assets and breaches its loss cap far more often than any Gaussian calculation predicts (Rockafellar u. Uryasev, 2000; Liu u. a., 2022).

1.1.3 A worked example: the cost of trusting the mean

The mechanism is easiest to see in a problem small enough to solve by inspection. Consider a single-bus dispatch model. A controllable resource of capacity x megawatts must cover an uncertain net demand ξ, and capacity is costly, so the planner solves

$$\underset{x \geq 0}{\text{minimize}}\; c\,x \quad \text{subject to} \quad \xi - x \leq 0. \tag{2}$$

Let net demand follow a moderately skewed lognormal law with mean $\bar{\xi} \approx 102\,\text{MW}$, a realistic shape for net load once renewable infeed is subtracted (Hodge u. Milligan, 2011). The mean-value method replaces ξ by $\bar{\xi}$ and returns $x^{\star}_{\text{mv}} = \bar{\xi}$: build exactly average capacity. Figure 1(a) shows what this decision looks like against the actual demand density. The constraint is violated on every day that demand exceeds its mean, and for this distribution that is 46% of all days. A plan that appears exact on paper fails almost every other day in operation.

The repair is equally easy to read off the figure. If the planner is willing to accept a shortfall on at most 5% of days, the correct decision is the 95% quantile of demand, $x = q_{0.95} \approx 134\,\text{MW}$, about 32% more capacity than the mean-value plan. Figure 1(b)

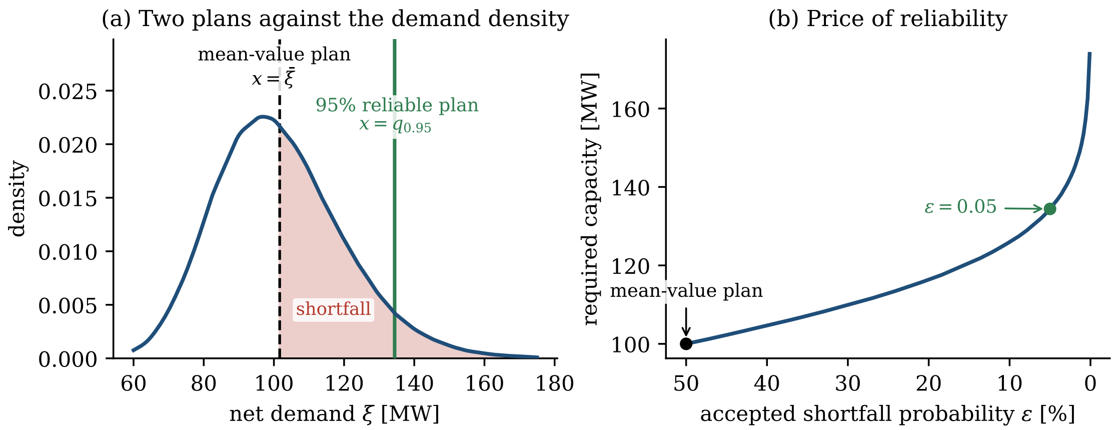

Figure 1: The mean-value fallacy in a one-dimensional dispatch problem with lognormal net demand (Monte Carlo with 2×10^5 samples). (a) The mean-value plan $x = \bar{\xi} \approx 102$ MW leaves the shaded shortfall region uncovered, so the constraint $\xi \leq x$ fails on 46% of days; covering demand with 95% reliability requires $x = q_{0.95} \approx 134$ MW. (b) Required capacity as a function of the accepted shortfall probability ε. Reliability is bought at an accelerating price as $\varepsilon \to 0$; the chance constraint makes this price explicit and lets the planner choose a point on the curve deliberately.

generalizes this observation into the central trade-off of the thesis: required capacity, and hence cost, rises steeply as the accepted violation probability ε shrinks, and rises without bound as $\varepsilon \to 0$ because the lognormal has unbounded support. Reliability is a commodity with a price, and the role of a properly formulated stochastic optimization problem is to make that price visible and chooseable rather than implicit and accidental.

1.1.4 A near-universal failure rate

One might hope that the 46% value is an artifact of the simple model. It is not. Figure 2 repeats the experiment for stylized versions of three of the domains above: dispatch under skewed demand, a landing-corridor constraint under bimodal thrust disturbance, and a portfolio loss cap under heavy-tailed (Student-t) returns. In every case the mean-value solution violates its constraint between 46% and 50% of the time, an order of magnitude beyond any tolerable design target. The violation frequency hovers near one half precisely because of the boundary mechanism described above: wherever the constraint is binding at the deterministic optimum, the median of the residual sits at or near zero, regardless of the application. The number is not a property of energy systems or of finance; it is a property of optimization itself.

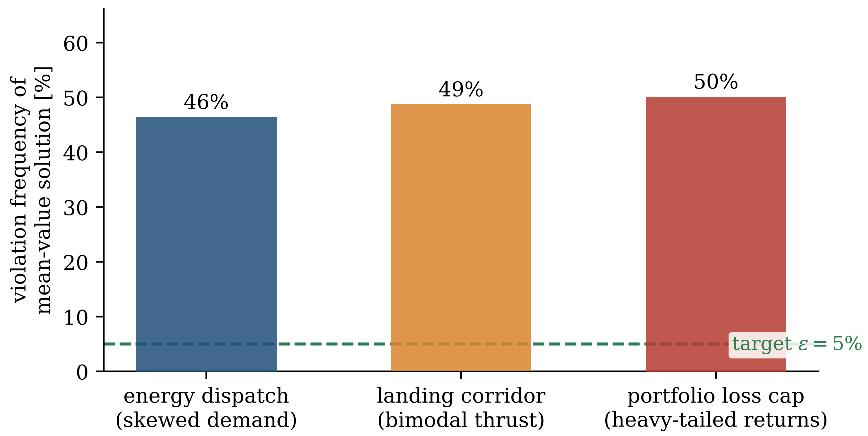

Figure 2: Empirical violation frequency of the mean-value solution in three stylized problems drawn from the application domains of this thesis (Monte Carlo estimates). A typical engineering target of $\varepsilon = 5\%$ is shown for scale. The near-universal value close to 50% reflects the boundary mechanism: optimizers drive binding constraints to the edge of feasibility, where half of all perturbations cause failure.

1.2 Two Extremes and the Gap Between Them

1.2.1 Ignoring uncertainty and worst-case robustness

Once the inadequacy of the mean-value approach is accepted, the conversation moves to what should replace it. Two answers dominate, and they sit at opposite ends of a spectrum.

The first answer is to ignore uncertainty entirely, which is the mean-value approach already dismantled in Section 1.1. Its defining property deserves restating in the language used for the rest of the thesis: the deterministic solution carries no reliability statement of any kind. Nothing in the formulation Eq. (1) bounds the probability of failure; Fig. 2 shows that this probability is typically near one half. Whatever reliability a deployed mean-value solution exhibits is an accident of safety margins added by hand, outside the optimization, by engineers who do not trust the model. Hand-tuned margins are a tacit admission that the optimization solved the wrong problem.

The second answer is worst-case robust optimization (Soyster, 1973; Ben-Tal u. a., 2009). The uncertain parameter is confined to an uncertainty set Ξ, and the constraint is enforced for every element of that set:

$$g(x, \xi) \le 0 \quad \text{for all } \xi \in \Xi. \tag{3}$$

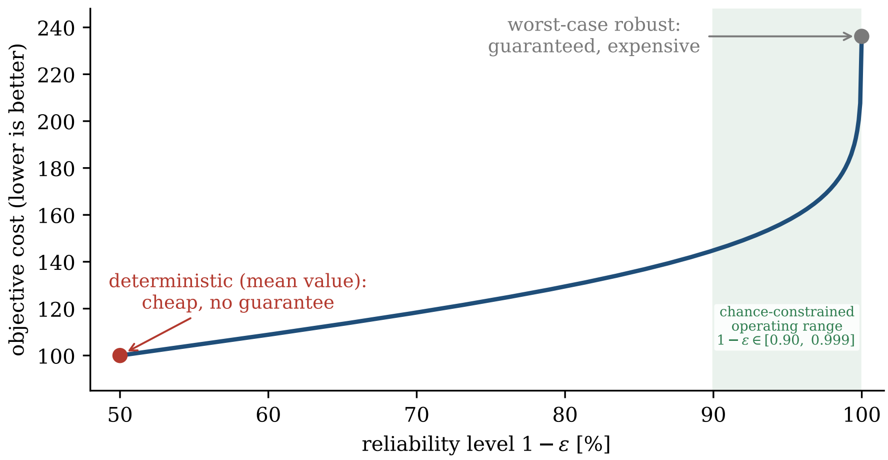

Figure 3: The reliability spectrum for the dispatch example of Fig. 1. The deterministic mean-value plan is cheap and offers no guarantee; the worst-case robust plan is guaranteed and expensive, and is not even well defined under unbounded support. Chance constraints parameterize the entire curve between the extremes: the decision maker selects a reliability level $1-\varepsilon$, and the optimization returns the cheapest decision that honors it.

Robust optimization is a mature and elegant theory; for many set geometries, Eq. (3) admits exact tractable reformulations, and when failure is truly never acceptable it is the correct model. Its weakness is the obverse of its strength: protection is purchased against every point of Ξ, including points that are vanishingly unlikely or physically implausible, and the objective pays for all of them. In the dispatch example, the demand distribution has unbounded support, so a literal worst-case formulation has no feasible solution at all; the set Ξ must be truncated by fiat, and the answer then depends sensitively on a truncation the data does not specify. Even with a defensible truncation, the protection level is an all-or-nothing property of the set, not a dial the decision maker controls. Empirically, robust solutions in power systems and aerospace routinely sacrifice a large fraction of attainable objective value to guard against scenarios whose combined probability is far below any operational concern (Ben-Tal u. a., 2009; Keil u. a., 2021).

1.2.2 When each extreme is the right tool

Fairness demands an acknowledgment before the middle ground is claimed: each extreme has a legitimate home. The mean-value approach is correct when uncertainty is genuinely negligible relative to the constraint margins, or when violations carry only a linear cost that the objective already prices in, as in inventory models where a shortfall is simply an expense. Worst-case robustness is correct when the uncertainty set is small, well characterized, and physically bounded, and when a single failure is catastrophic and irreversible: structural safety factors live here for good reason. The problems this thesis targets sit in neither home. Their uncertainties are large, data-described, and unbounded or effectively so, and their failures are costly but tolerable at a small, explicit rate. A lander program accepts a quantified risk per descent; a grid code tolerates a bounded loss-of-load expectation per year. For this regime, neither extreme expresses what the decision maker actually wants.

1.2.3 Chance constraints as the unifying middle ground

Between a method with no guarantee and a method that guarantees too much lies the formulation this thesis is about. A chance constraint, introduced by Charnes u. Cooper (1959) and developed into a full branch of stochastic programming by Prékopa (1995), requires the constraint to hold with a prescribed probability:

$$\mathbb{P}\big(g(x,\xi) \leq 0\big) \geq 1-\varepsilon. \tag{4}$$

The risk tolerance $\varepsilon \in (0,1)$ is small, typically between 0.001 and 0.1 depending on the stakes, and $1-\varepsilon$ is the reliability level. Figure 3 places the three formulations on a single axis for the dispatch example. The chance constraint is not a compromise between the extremes; it is a strict generalization of both. As $\varepsilon \to 0$ it recovers worst-case protection over the support of ξ, and at $\varepsilon = 0.5$ it roughly reproduces the mean-value behavior for symmetric residuals. Every intermediate point is available, and each carries an explicit, auditable meaning: the system fails at most a fraction ε of the time. This is exactly the language in which engineering requirements are written. Grid codes prescribe loss-of-load expectations, flight rules prescribe acceptable failure probabilities per descent, and risk mandates prescribe loss probabilities per quarter. The chance constraint is the optimization-native translation of a reliability requirement, which is why it is the right framing for most engineering and operational problems, and why the difficulty of solving it, the subject of **??**, has remained an active research frontier for over six decades.

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Erlangen, den June 14, 2026 Hierher die Unterschrift

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